

# Improvements in Exceptional Learning

## *Detailed Code Review Report (Category Wise)*

### 1. Code Structure & Navigation Optimization

- Organized imports and removed unused imports.
- Centralized route configuration using routes array.
- Improved scalability and maintainability.
- Added key prop for mapped navigation screens.

### 2. Conditional Rendering Optimization

- Removed unnecessary ternary operators.
- Replaced with short-circuit conditional rendering.
- Improved JSX readability and cleanliness.

### 3. Performance Optimization (Memoization & Re-renders)

- Used useCallback for prop functions.
- Used useMemo for optimized style calculations.
- Reduced unnecessary re-renders in FlatList.
- Added accessibility labels for better UX.

### 4. API & Security Improvements

- Replaced AsyncStorage token storage with encrypted MMKV storage.
- Standardized Authorization header usage.
- Improved authentication security for production.

### 5. TypeScript Strict Typing (API Layer)

- Created proper interfaces for API payload and response.
- Removed usage of 'any' type.

- Improved production stability and crash prevention.

## **6. Component Optimization (React.memo)**

- Wrapped list items in React.memo.
- Added keyExtractor in FlatList.
- Improved rendering performance.

## **7. Styling Standardization**

- Moved inline styles to StyleSheet.create().
- Improved performance and style reusability.
- Centralized design consistency.

## **8. Dependency & Production Configuration**

- Updated TypeScript to latest stable version.
- Removed yarn from dependencies.
- Pinned critical dependency versions.
- Added engines configuration for Node and NPM.

## **9. Production Readiness Enhancements**

- Environment-based configuration setup (Dev, Staging, Prod).
- Integrated Sentry error tracking.
- Improved accessibility compliance.

## **Overall Impact**

- Cleaner and scalable architecture.
- Improved application performance.
- Enhanced security and encrypted token storage.
- Production-ready TypeScript implementation.
- Industry-standard best practices applied.